Project Timeline & Resource Plan

DEPI · Project Planning & Management · Railway Ticket Sales & Journey Performance · Oct 22 – Dec 4, 2026

44	9	9	6	5
Days in total	Phases	Milestones	Team members	Tools

2.1 Project timeline

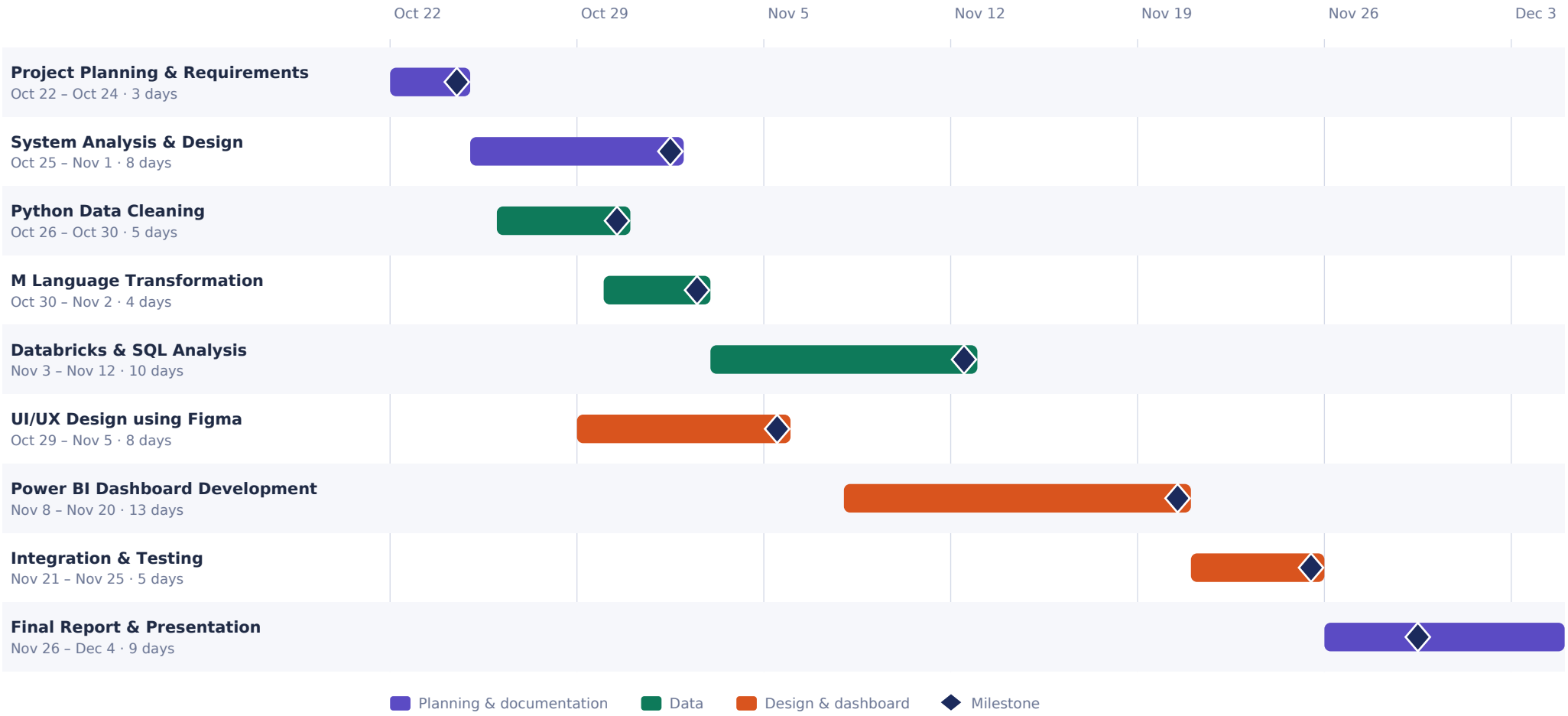
#	Phase	Start	End	Length	Main activities
1	**Project Planning & Requirements**	Oct 22	Oct 24	3 d	Define project objectives, scope, requirements, and team responsibilities.
2	**System Analysis & Design**	Oct 25	Nov 1	8 d	Understand the dataset, identify analytical questions, define KPIs, and plan the dashboard structure.
3	**Python Data Cleaning**	Oct 26	Oct 30	5 d	Inspect data types, missing values, duplicates, and inconsistent categories; prepare the cleaned dataset.
4	**M Language Transformation**	Oct 30	Nov 2	4 d	Perform data transformation using Power Query (M Language) to clean, structure, and prepare data for analysis.
5	**Databricks & SQL Analysis**	Nov 3	Nov 12	10 d	Prepare database tables and analyze ticket sales, routes, journey status, delays, and refund requests. The SQL work is reviewed before the milestone is closed.
6	**UI/UX Design using Figma**	Oct 29	Nov 5	8 d	Create wireframes, define dashboard layouts, and design the user interface.
7	**Power BI Dashboard Development**	Nov 8	Nov 20	13 d	Develop interactive dashboards, KPIs, charts, and slicers based on the analysis results.
8	**Integration & Testing**	Nov 21	Nov 25	5 d	Validate analytical results, test dashboard interactions, and fix errors.
9	**Final Report & Presentation**	Nov 26	Dec 4	9 d	Complete project documentation, prepare presentation slides, and finalize deliverables.

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2.2 Gantt chart

The chart shows the schedule and the overlap between tasks. Parallel activities let team members work efficiently while keeping data preparation, analysis, and visualization coordinated.



Optional (stretch goal): a baseline machine learning model is not part of the schedule above. The team will decide after the SQL analysis is completed whether to build it, depending on the remaining time and on how the core work is going. The project is complete without it.

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2.3 Project milestones

Milestones are checkpoints used to track progress and confirm that the required deliverables are ready.

Milestone	Target date	Expected deliverable
Project Plan Completed	Oct 24	Approved project scope, objectives, requirements, and task assignments.
Cleaned Dataset Ready	Oct 30	Cleaned and standardized dataset prepared using Python, with documented data-quality decisions.
System Analysis & Design Completed	Nov 1	Dataset assessment, analytical questions, KPI definitions, and initial dashboard plan.
Transformed Data with M Language	Nov 2	Structured, prepared, and cleaned data using Power Query (M Language) for analysis.
Figma Prototype Completed	Nov 5	Dashboard wireframes and UI/UX prototype.
SQL Analysis Completed	Nov 12	Database tables, SQL queries, and analytical results for sales, routes, delays, and refunds, reviewed by a second team member.
Power BI Dashboard Completed	Nov 20	Interactive dashboard containing KPIs, charts, and slicers.
Testing and Validation Completed	Nov 25	Verified analytical results, tested dashboard functionality, and resolved identified issues.
Final Project Submission	Nov 29	Final report, presentation slides, dashboard, and supporting project files.
Optional: ML Model Decision	After Nov 12	The team decides whether to build a baseline ML model. Only if there is enough time and the core work is on track.

2.4 Deliverables and completion criteria

The project is complete when these are ready:

Cleaned Dataset
Data quality issues have been investigated, and cleaning decisions are documented.

Transformed Data
Data has been transformed with Power Query (M Language) and is ready for analysis.

SQL Analysis
Queries and results address the project's analytical questions and have been reviewed.

Figma Prototype
The dashboard layout and user interface are clearly defined.

Power BI Dashboard
The dashboard presents the required KPIs and visualizations, with functioning filters.

Final Documentation
The report explains the project objectives, methodology, findings, and limitations.

Final Presentation
The team can demonstrate the dashboard and explain the key findings.

ML Model (optional)
Only if the team decides to build it. Not required for project completion.

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2.5 Resource allocation

Resource allocation defines how the available team members, tools, and datasets are assigned to the project activities to ensure efficient execution and timely completion.

ME — Moustafa Elsayed
Data Cleaning

Inspect data quality, handle missing values appropriately, remove or investigate duplicates, standardize categories, and prepare the cleaned dataset.

Python · Pandas · railway.csv · railway_data_dictionary.csv

OS — Omar Saad
M Language Transformation, Database Preparation & SQL Review

Transform the cleaned data using Power Query (M Language), set up the database environment, organize tables, prepare the data for querying, and review the SQL analysis.

Power Query (M) · Databricks · SQL

NA — Nada Ahmed
Data Analysis

Analyze ticket sales, ticket types, routes, journey status, delay reasons, and refund requests. SQL work is reviewed by Omar Saad.

Databricks · SQL

SA — Saja Ahmed
UI/UX Design

Design dashboard wireframes, organize the layout, and define the visual structure and user experience.

Figma

JM — Jana Mohamed
Dashboard Development

Build dashboard pages, KPIs, charts, and interactive filters.

Power BI

HE — Habiba Eslam
Visualization & Validation

Refine dashboard layouts, improve visual consistency, test interactions, and compare dashboard results with SQL outputs.

Power BI · Figma · SQL results

All team members: define requirements, review findings, validate deliverables, prepare the final report, and present the project.

Data resources

- `railway.csv`: the main dataset (ticket transactions, journey details, prices, journey statuses, delay reasons, refund requests).
- `railway_data_dictionary.csv`: the reference file describing the dataset fields.

Technical resources

- Python: data inspection, cleaning, and preparation.
- Power Query (M Language): data transformation and preparation for analysis.
- Databricks and SQL: database preparation and analytical queries.
- Figma: UI/UX design and dashboard prototyping.
- Power BI: interactive visualization and dashboard development.